

SE 216 – SOFTWARE PROJECT MANAGEMENT

SOFTWARE MEASUREMENTS DOCUMENT

PROJECT NAME: Happy Tails

GROUP NUMBER and MEMBERS: Semih Öztürk, Mirza Kerem Kılıç, Melike Gülmüne Kırpat, Özge Öztürk, Hüseyin Günççek, Arda Özkaranfil, İdil Buse Zengin

Questions to identify measurements:

- Did the project progress according to the planned schedule, specifically for core features like "Animal Listing" and "Adoption Appointment"?
- How much effort (staff-hours) was spent on complex tasks?
- How much effort was spent on the integration of external products (Open AI, Agora, Stripe)?
- How many defects were identified during the "Test Before Deployment" phase in Sprint 4?
- How effective were the security measures like "Encode User Data with SHA-256"?
- Does the software meet the expectations of the stakeholders?

Identified measurements:

- Planned sprint backlogs / Completed sprint backlogs
- Planned task deadlines / Met task deadlines
- Planned time to complete tasks / Actual time to complete tasks
- Defects encountered / Critical defects
- Total test cases / Passed test cases
- Current user requirements / Implemented requirements
- Successful responses from OpenAI and Agora / Total requests
- Percentage of code covered by automated tests

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Measurement storage and collection:

What: Sprint Backlogs and Tasks

- When: At the end of each sprint
- Format: Gantt chart tasks
- How: Keeping a record of each completed task

What: Defects and Test Cases

- When: Test Before Deployment phase
- Format: Integer counts in defect log
- How: Keeping a shared record of test results

What: Current Requirements

- When: Before Deployment
- Format: Updated list of requirements
- How: Analyzing with the stakeholders

What: API Performance and Reliability

- When: During Integration Testing and Production
- Format: Log files and response time milliseconds
- How: Monitoring tools for Stripe, Agora, and OpenAI calls

Measurement Type	Description	Example Measurements
Schedule	Compares the planned finish date with the real finish date	UI Design: Planned Apr 7, Finished Apr 9
Effort	Total hours spent on development tasks each week.	Sprint 1: Planned 80 hours total.
Quality	Total number of bugs identified during a specific sprint.	4 defects found in Medical Management.
Quality	Number of passed test cases.	8 tests were passed in the API implementation out of 10.
Productivity	Measures the amount of work completed in a sprint.	Velocity: 45 Story Points completed in Sprint 3.

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Reliability	Success rate of third-party API integrations.	Stripe Integration: 99% successful payment transactions.
Performance	Speed of API features like Stripe, Agora, and OpenAI	Response Time: Page loads in 1.8 seconds.